

Generation Interconnection System Impact Study Report

For Queue Project Q22-2064

Sycamore Cogen 100 kV

March 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2064. The Interconnection Customer proposes a 94.1 MW battery energy storage facility interconnecting at the Sycamore Cogen 100 kV substation in the Illinois area. The request was received on April 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 719 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,653,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Laning to Dyckesville 100kV
Contingency: Sand Ridge to Castle Afb 100kV
Loading: 194.2% (Rating: 133 MVA)
DFAX: 22.2%
Pre-project loading: 178.6%
2. Facility: Lawrence Hill Substation to Terryville 100kV
Contingency: S Boston to Blythe 230kV line
Loading: 190.6% (Rating: 133 MVA)
DFAX: 30.5%
Pre-project loading: 169.0%
3. Facility: Terryville to Sycamore Cogen 100kV line
Contingency: S Boston to Blythe 230kV Line
Loading: 173.4% (Rating: 137 MVA)
DFAX: 30.5%
Pre-project loading: 152.4%
4. Facility: Sand Ridge Substation to Castle Afb 100kV
Contingency: Lansing to Dyckesville 100kV L
Loading: 168.9% (Rating: 134 MVA)
DFAX: 18.4%
Pre-project loading: 155.9%
5. Facility: North Vly to West Danville 15 Substation 100kV Line
Contingency: Weyerhaeuser Co to North Vly 100kV L
Loading: 161.4% (Rating: 129 MVA)
DFAX: 8.1%
Pre-project loading: 155.5%
6. Facility: North Valley to Lawrence Hill 100kV
Contingency: Sweatt to Tinley Park 100kV
Loading: 148.8% (Rating: 209 MVA)

DFAX: 11.2%

Pre-project loading: 143.8%

7. Facility: Sweatt to Lawrence Hill 100kV

Contingency: Tinley Park to Sweatt 100kV

Loading: 141.0% (Rating: 229 MVA)

DFAX: 5.6%

Pre-project loading: 138.7%

8. Facility: Shoreham Power Sta to Castle Afb Substation 100kV

Contingency: Dyckesville Substation to East Jackson 1&2 100kV L

Loading: 126.2% (Rating: 137 MVA)

DFAX: 15.2%

Pre-project loading: 115.7%

9. Facility: North Vly to Weyerhaeuser Co 100kV

Contingency: W Danville 15 to N Valley 100kV

Loading: 121.9% (Rating: 131 MVA)

DFAX: 8.4%

Pre-project loading: 115.9%

10. Facility: Castle Afb to Dyckesville 100kV Line

Contingency: Cstle Afb to Sand Ridge 100kV Line

Loading: 119.3% (Rating: 135 MVA)

DFAX: 7.1%

Pre-project loading: 114.3%

11. Facility: Franz Holmes to Sand Ridge 100kV

Contingency: Lansing to Dyckesville 100kV

Loading: 118.3% (Rating: 138 MVA)

DFAX: 14.0%

Pre-project loading: 108.7%

12. Facility: Dyckesville to East Jackson 1&2 100kV

Contingency: Castle Afb to Shoreham Power Station 100kV

Loading: 111.7% (Rating: 154 MVA)

DFAX: 20.4%

Pre-project loading: 99.2%

13. Facility: Shoreham Power Sta to Outlook Substation 100kV Line

Contingency: South Holland to Shoreham Power Station 100kV Line

Loading: 108.1% (Rating: 118 MVA)

DFAX: 9.5%

Pre-project loading: 100.5%

14. Facility: Lincoln Substation to Blythe 230kV

Contingency: South Boston to Blythe 230kV Line

Loading: 106.5% (Rating: 617 MVA)

DFAX: 6.1%

Pre-project loading: 105.6%

15. Facility: Lansing to Ucon 100kV

Contingency: Lansing to Ucon 100kV #2

Loading: 105.3% (Rating: 146 MVA)

DFAX: 13.7%

Pre-project loading: 96.5%

16. Facility: Lansing to Ucon 100kV Line 2

Contingency: Lansing Line to Ucon 100kV Ckt 1

Loading: 105.3% (Rating: 146 MVA)

DFAX: 13.7%

Pre-project loading: 96.5%

17. Facility: Canadys Steam to Lawrence Hill 100kV

Contingency: Canadys Steam to Lawrence Hill 100kV Circuit 2

Loading: 103.5% (Rating: 231 MVA)

DFAX: 6.3%

Pre-project loading: 101.0%

18. Facility: Canadys Steam to Lawrence Hill 100kV

Contingency: Canadys Steam to Lawrence Hill 100kV #1

Loading: 103.5% (Rating: 231 MVA)

DFAX: 6.3%

Pre-project loading: 101.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lansing to Dyckesville 100kV line	Rebuild line with higher-capacity conductor	\$1,288,000	17% (\$219,000)
Lawrence Hill Substation to Terryville 100kV	Rebuild line with higher-capacity conductor	\$1,865,000	15% (\$281,000)
Terryville to Sycamore Cogen 100kV line	Rebuild line with higher-capacity conductor	\$2,692,000	15% (\$406,000)
Sand Ridge Substation to Castle Afb 100kV	Rebuild line with higher-capacity conductor	\$716,000	16% (\$112,000)
North Vly to West Danville 15 Substation 100kV Line	Rebuild line with higher-capacity conductor	\$3,349,000	5% (\$164,000)
North Valley to Lawrence Hill 100kV	Rebuild line with higher-capacity conductor	\$2,374,000	4% (\$86,000)
Sweatt to Lawrence Hill 100kV line	Rebuild line with higher-capacity conductor	\$3,911,000	2% (\$68,000)
Shoreham Power Sta to Castle Afb 100kV	Rebuild line with higher-capacity conductor	\$3,341,000	14% (\$461,000)
North Vly Substation to Weyerhaeuser Co 100kV	Rebuild line with higher-capacity conductor	\$3,364,000	6% (\$198,000)
Castle Afb to Dyckesville 100kV Line	Rebuild line with higher-capacity conductor	\$1,675,000	13% (\$215,000)
Sand Ridge to Franz Holmes 100kV Line	Rebuild line with higher-capacity conductor	\$1,068,000	31% (\$329,000)
Dyckesville to E Jackson 1&2 100kV	Rebuild line with higher-capacity conductor	\$2,052,000	24% (\$490,000)
Outlook to Shoreham Power Sta 100kV	Reconductor line	\$732,000	12% (\$89,000)
Blythe to Lincoln 230kV	Reconductor line	\$5,290,000	1% (\$45,000)
Ucon Substation to Lansing 100kV	Reconductor line	\$564,000	19% (\$108,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lansing to Ucon 100kV Line 2	Reconductor line	\$495,000	19% (\$94,000)
Canadys Steam to Lawrence Hill 100kV	Reconductor line	\$2,139,000	7% (\$145,000)
Canadys Steam to Lawrence Hill 100kV	Reconductor line	\$2,108,000	7% (\$143,000)
	Total allocated to project		\$3,653,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.